

OrchiZen: Hybrid Integrated Smart Farming System for Orchid Plantations

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Abstract— OrchiZen is a hybrid integrated smart farming system designed for orchid cultivation, leveraging Machine Learning (ML) and Internet of Things (IoT) technologies to address key horticultural challenges, including irrigation, disease treatment, choice of species, lighting, and nutrients. The OrchiZen has smart irrigation advisory, species recommendation, Ultraviolet (UV) based disease treatment, light optimization, and fertilizer advisory. The priorities are given to specific species such as Dendrobium, Vanda, and Phalaenopsis. The realities of telemonitoring, data processing, and forecasting increase organizational productivity and contribute to better environmental management. The outcomes illustrate that existing modern technologies can enhance the output and ecology of the orchid production to a significant extent, redefining the conventional technologies.

Keywords— orchid, mist irrigation, UV-C treatment, fertilizer advisory, sustainable agriculture

I. INTRODUCTION

Orchids refer to plants that are in the Orchidaceae family and are some of the most diverse and distributed flowering plants across the world [1]. These are plants that are perennial in nature and favored for their structural flexibility, ornate flowers, and incredible tolerance to frequently, changing ecosystems [1]. Orchids have a high ornamental importance and are used around the world in the floriculture business. Fragrance characteristics and appearance make Dendrobium, Vanda, and Phalaenopsis beautiful and trendy, and because these species remain blooming for a long time [2]. Orchids are grown mainly for their ornamental value and given their popularity in the local and international markets, they are economically vital [1], [2].

The orchid management tends to have several realistic problems. High levels of environmental control are essential for these plants, especially environmental variables such as temperature, humidity, and light intensity [1]. Irrigation is critical since when done inappropriately; it has a negative influence on plant health through either inadequate or excess water [1]. Choosing species that can grow well in certain conditions adds to the challenges of cultivation since the wrong species can lead to improved performance [2], [3]. Nutrient needs change with growth stages, making it necessary to have the right recommendation on fertilizers that will feed the crops [3]. Fungal diseases are prevalent in orchids, especially when left uncontrolled as it can spell doom

to the growers [4]. Light requirements are similar to heat as well; they also affect photosynthesis and growth in cases of inadequate or excessive light [1], [2]. It was observed that in a conventional farming system, the crop necessity is not adequately met, farm yield reduces, and the farmer incurs a loss [1]. To overcome these challenges, this study proposes OrchiZen: The study enquires with ML, IoT and data intelligence with the aim of forging a sustainable and efficient system for growing orchids. Its preliminary focuses on Dendrobium, Vanda, and Phalaenopsis species to enhance control of microclimatic conditions, estimate the yield, disease diagnostic tools, pest control, and nutrient supply.

The smart irrigation system in the OrchiZen Delivery System guarantees water delivery based on environmental factors such as humidity, temperature, and light intensity. It retails resource wastage and maintains favorable conditions for plant health. The construction of the system is also fairly versatile and can be expanded and refined to meet and adjust to different sorter requirements particular to the different types of orchids. To meet these needs by controlling fruit development and maturation, which possess a negative impact on yields and income of the grower. An orchid species recommendation system uses IoT sensors and Artificial Neural Networks (ANN) for species recommendation in consonance with the existing environmental conditions. Along with the usage of historical information and current parameters, this system also minimizes the number of mismatches, thereby promoting greater yields and rational use of available resources.

Light optimization within OrchiZen is based on ML algorithms and IoT technologies that allow the selecting the appropriate light level for each species. It guarantees the right amount of light to the orchids which increases flowering and growth. Disease management is realized through the use of image-based ML to identify early-stage fungal infection. The system allows to carry out particular dissection by the UV-C treatment without harm to the plants themselves, allowing to prevent disease spread while keeping plants healthy. The fertilizer recommendation system is similar to the former and provides the nutrient requirements based on real time monitoring and mining of data at different stages. It focuses on delivering fertilizers accurately according to the growth stage resulting in efficient use of fertilizers and reduced costs, thereby promoting sustainable production of orchids.

Consequently, OrchZen provides an innovative solution for the dilemma in orchid cultivation by addressing productivity, sustainability, and profitability of the protected environment.

II. LITERATURE REVIEW

Considerable number of studies have emphasized net irrigation through plant weight for assessment of moisture requirements with overhead sprinkling [5]. Although it is practical for large scale applications, it does not have the localized flow control that is required to alter according to the different circumstances of the different orchid varieties. Nevertheless, this study uses an overhead mist irrigation system which is most appropriate with respect to the root and aerial part of the orchids, lowering tendencies of over irrigation. Another other study conducted analyzed IoT based programmed watering using fuzzy logic [6]. Nevertheless, it does not consider the relationships between the four factors, humidity, temperature and light, which are important for flexible irrigation in changing conditions. Similarly, the use of soil moisture sensors in epiphytic orchids was an issue of minimal soil contact [7]. The systematic watering for greenhouses is useful where the exterior conditions do not change frequently, but they are inapplicable under unpredictable weather conditions [8].

The Study on the distribution and diversity of orchids in Central Europe has shown how important ecological elements like humidity, temperature, sun radiation, and land cover are. Experts found that temperature and humidity are important factors for orchid dispersal using ecological niche modeling [9]. According to the effects of temperature, humidity, and light intensity on orchid population dynamics and adaptive responses to environmental changes were investigated by statistical models and long-term ecological investigations. Both studies showed shortcomings in capturing the intricacy of environmental interactions, highlighting the necessity of fine-grained, real-time monitoring systems to improve forecasts, maximize farming for specific species, and assist focused conservation efforts [10].

The use of UV-C light for preventing fungal pathogens in agriculture was studied, demonstrating a great reduction in disease by disrupting fungal reproduction through DNA treatment of targeted effects. The system with reflective linings to increase light efficiency and safety has proven to be an excellent alternative to chemical-based methods and can be installed in various agricultural environments [13]. Hydroponic systems also were subjected to IoT-based grow light automation, where changing wavelengths and intensities in real time increased plant growth parameters while reducing resource use, creating near-perfect conditions for raising high-value crops [12]. Explainable deep neural networks were employed to recognize orchid species accurately so that researchers can shape cultivation techniques [14]. Machine learning is employed for Orchid Virus Disease identification with U-Net and random forest classifiers. They identify *Odontoglossum Ringspot Virus* and *Cymbidium Mosaic Virus* with an accuracy rate of 91.8%. This proves the capability of advanced texture analysis and image processing for early detection and intervention [11].

Orchids need five essential nutrients for growth: nitrogen, phosphorus, potassium, calcium, and magnesium, which they absorb through the roots' velamen layer [15]. Like any other plant, variations in calcium and magnesium content are common [15]. Alongside these nutrients, the optimum

balance of iron, zinc, and boron facilitates plant health and flowering [15]. Each species has its own specific nutrient requirements; for instance, the NPK ratio for *Dendrobium*, *Vanda*, and *Phalaenopsis* orchids is very crucial for growth. For instance, *Phalaenopsis* orchids require that the proportions be modified as per their minimum generation time (4:0.8) or multiplication rate is enhanced [15]. With the aid of Plant Growth Promoting Rhizobacteria (PGPR), Hybrid Integrated Smart Farming helps in nutrient absorption, root growth promotion, and making the environment sustainable [17]. AI-driven IoT greenhouses with over 98.6% accuracy in disease detection and crop management, through mechanisms like CNN and SVM, carve an elite market for orchids in Taiwan [16].

Unlike traditional systems that merely sprinkle overhead, the smart irrigation system from OrchZen adjusts the water delivery dynamically, considering environmental factors in real-time: humidity, temperature, or light intensity-wasting less of the resource [5]. Unlike these adaptable systems, conventional systems are hardwired into a schedule and may either overwater or allow the water to drain off, thus losing efficiency. Another feature to discriminate OrchZen from traditionally fungal management systems would be that it integrates early disease detection through image-based machine learning rather than depending on chemical treatment, which may be damaging to the plants and more reactive. It has been established that UV-C treatment by OrchZen with reflective linings for increased light efficiency is a safer and more effective alternative to chemical methods [13]. While OrchZen affords an accuracy rate of 91.8% in detecting fungal diseases, traditional methods are based on late intervention [11].

III. PROPOSED SYSTEM

The system overview provides a concise insight into the essential components namely automated irrigation mist machine, selected species under environmental conditions, orchid UV disease treatment, and light optimization and fertilizer identification.

A. Automated Irrigation Mist Machine

The new developed system suggests an overhead mist irrigation system that's specific for the *Dendrobium*, *Vanda* and *Phalaenopsis* Orchids as the problem presented does not interest simple and stable aquatic environments. In contrast to the fixed techniques, this component applies an algorithmic technique that can enable irrigation to change daily as required by the environmental factors of humidity, temperature, and light intensity all of which are correlated and vital for the growth of an orchid. Through the regular monitoring of these parameters, it guarantees the proper distribution of water under fine misting to avoid over irrigation or under irrigation. This process refines water efficiency whilst establishing ideal climate conditions for growth of the specific selected orchid species to improve on their growth ad sustainability.

B. Species Selection under Environmental Conditions for the Selected Area

The Species Selection module is a decision support system designed to recommend the optimal orchid species for

specific climate conditions. By analyzing environmental factors such as climate, temperature, and sunlight, it provides precise recommendations tailored to the growth requirements of various orchid species. The module incorporates a detailed knowledge base that aligns environmental suitability with the biological and economic viability of different orchid stocks, ensuring sustainable and cost-effective solutions. By offering real-time insights and fostering informed decision-making, this system supports farmers in selecting species best suited for their local conditions, promoting efficient and sustainable orchid cultivation practices.

C. Orchid UV Disease Treatment & Light Optimization

The system will provide an integrated approach for the optimization of lighting conditions and the management of fungal diseases in orchid farming. The species-specific care of orchids like *Phalaenopsis*, *Vanda*, and *Dendrobium* is made possible by this system, the intensity of grow light changes dynamically according to ambient conditions. The proposed system treats fungal infections with precision, using targeted UV-C light exposure that stimulates plant recovery and is safe for the surrounding plants and users. It integrates intelligent disease detection, and light optimization mechanisms that will facilitate orchid cultivation to create an environment for healthy growing and efficient resource utilization.

D. Recommend the Suitable Fertilizers according to the Plant Growth

Concerning fertilizers, the proposed system includes such products as NPK, Calcium Nitrate and Potassium Sulfate and the proposed quantities depend on the plant development stages namely baby, child, young, mature, adult and the features of the plant's leaves, roots, flowers, and stems. Using a considerable set of intelligent algorithms has emphasized a plant age gap and based on the analysis, provides a precise nutrient requirement that may not lead to over-fertilization or nutrient deficiency. Recommendations change dynamically with the features of plants and environmental conditions to give way to healthy growth and resource efficiency. With this, the smart farming system will bring successful orchid cultivation in terms of optimized growth, improved quality of yield, and sustainability.

IV. METHODOLOGY

This study focused on enhancing the concept and design of the hybrid integrated smart farming system for the orchid plantations focusing on four main areas a mist irrigation machine, an environmental species selection system for a particular area, an orchid UV disease treatment and light control system, and an orchid fertilizer recommendation system according to plant growth.

A. Automated Irrigation Mist Machine

Mist system is installed to enhance the watering of the three chosen orchid types belonging to *Dendrobium*, *Vanda* and *Phalaenopsis*. It incorporates a sophisticated microcontroller which controls the rate of irrigation depending on the prevailing environmental conditions of light, humidity, and temperature, all of which are fundamental and related factors influencing the water requirements of orchids. The algorithm was incorporated to manage the

interdependencies in these measures to provide accurate and dynamic irrigation.

The operations in the system are managed through the Espressif Systems Processor 32-bit (ESP32) microcontroller which is considered its firmware. Digital Humidity and Temperature sensor (DHT11) employed for detect humidity and temperature values and Light Dependent Resistor (LDR) employed for detect light intensity. They supply the needed information to the algorithm which is demonstrated further in the context of the account. Further, Switched Mode Power Supply (SMPS) module employed for convert Alternating Current (AC) to Direct Current (DC) for the smooth running of the system's electronics. And the mist generated using 12V motor which is using the SPMS DC current.

The watering mechanism selected was overhead cone shape mist irrigation because of equal distribution of water and ability to retain the microclimate required for good growth of the orchids. The incorporation of real time environmental sensing with an efficient algorithm and effective hardware component makes the system ideal to dynamically changing environment.

B. Orchid UV Disease Treatment and Light Optimization

The proposed UV light treatment and light optimization system combines IoT and ML technology to provide *Dendrobium*, *Vanda*, and *Phalaenopsis* orchids with accurate disease management and environmental control solutions. The system uses two You Only Look Once Version 5 (YOLOv5) based machine learning models to identify fungal diseases and classify orchid species. After collecting and preprocessing leaf photos for model training, *Phalaenopsis*, *Vanda*, and *Dendrobium* detection. A YOLOv5 model was trained to use image analysis to detect fungal illnesses like *Phytophthora*, *Anthracnose*, and *Fusarium*. The technology ensures focused intervention without manual supervision by initiating UV-C treatment upon illness diagnosis. Previous studies have confirmed that deep learning models are effective for classifying plants and extracting explainable features [11].

The light optimization system uses AC dimmers and LDR sensors to dynamically control the growing light intensity. Based on predetermined limits for each species of orchid, the system classifies ambient light levels into low, medium, and high categories in real-time. An ESP32 microcontroller processes real-time data from LDR sensors and uses AC dimmers to modify light intensity to suit species-specific needs. To minimize electricity use, the system maximizes illumination during low ambient circumstances and decreases intensity when external light increases. According to studies [12], this automated optimization procedure reduced energy consumption by 25% when compared to manual techniques. Fungal infections identified by the YOLOv5 model were addressed by UV-C light treatment at a controlled intensity of 30 J/cm² in order to manage the disease. To reduce phototoxic effects on the plants, the device used a 275 nm UV-C lamp to produce pulsed UV-C light exposures. The study has demonstrated that pulsed UV-C treatments efficiently eradicate fungal infections while maintaining plant safety, as demonstrated by studies like those conducted by McGill University [13]. To shield users and surrounding plants from dangerous radiation, the UV treatment was carried out in a UV safety box, which

offered a controlled atmosphere. To further mitigate fungal infections and encourage plant recovery, complementary fungicide recommendations were also incorporated into an integrated disease management strategy.

A web-based interface running on the ESP32 microcontroller allows real-time monitoring and control of the complete system. Important parameters like UV-C treatment status, disease alerts, current growth light intensity and ambient light levels were shown on the interface. This improved the system's usability and scalability for orchid farming by enabling remote monitoring and control. The system's capacity to efficiently control fungal infections and optimize illumination was validated by validation tests. The effectiveness of UV-C in controlling illness was demonstrated by the 70% improvement in plant health that was seen after just one cycle of treatment.

C. Species Recommendation

The Orchid Species Recommendation System provides optimal orchid species recommendations based on environmental conditions. IoT sensors, including a hygrometer, thermistor, and photometer, monitor key factors like humidity, temperature, and light intensity. Data is collected twice daily for seven days to capture daily and seasonal variations, ensuring accurate analysis. After collection, data preprocessing involves averaging to minimize anomalies, normalization for uniform scaling, and preparation for the ANN. The ANN analyzes the processed data to recommend suitable orchid species namely Phalaenopsis, Dendrobium, or Vanda. Training the model uses historical data paired with corresponding species, employing a feed-forward network and backpropagation to reduce errors. The system is validated with unseen data to ensure reliability.

Integrated into the Hybrid Integrated Smart Farming System, the system uses secure protocols to process real-time sensor data. Recommendations combine with insights from other components, such as fertilizer and irrigation, offering a comprehensive farming strategy. Farmers receive user-friendly, actionable recommendations to enhance orchid cultivation. The system continuously learns, incorporating farmer feedback and newly recorded data to improve accuracy and adapt to diverse conditions. Leveraging IoT and AI, it provides sustainable, optimized solutions for orchid farming.

D. Recommend the Suitable Fertilizers According to the Plant Growth

The proposed system detects suitable fertilizers based on the identified these features including the age gap and species Dendrobium, Vanda, and Phalaenopsis. The use of YOLOv11 provides three photos of the plant, and the machine measures particular traits including the number of leaves, roots, flowers, and stems [16]. The characteristics are further processed using a machine learning algorithm to estimate the age gap of the plant [15]. The derived recommendations specify the type, amount, and time of fertilizer application. Fertilizer type is species-specific and growth-stage-dependent, Dendrobium requires nitrogenous fertigation during the vegetative phase and phosphatic fertigation during the flowering phase, while Vanda requires high nitrogen at the early stage and potash later [15]. In addition, the size and character of the plant are considered, and the fertilizers are proportionately adjusted based on the size example, 1 g/leaf for young plants. A flexible fertilization schedule is proposed with weekly applications during strong growth periods and biweekly to monthly during

maintenance periods. The advice is provided via a mobile app, created with react native for the front-end interface and flask-python for the backend infrastructure, in order to ensure that the farmers get personalized recommendations. This system has been efficient in reducing fertilizer usage and increasing plant health, substantially contributing to growing orchids sustainably [17].

Understanding of the system structure the visual representation of the proposed system can be shown below.

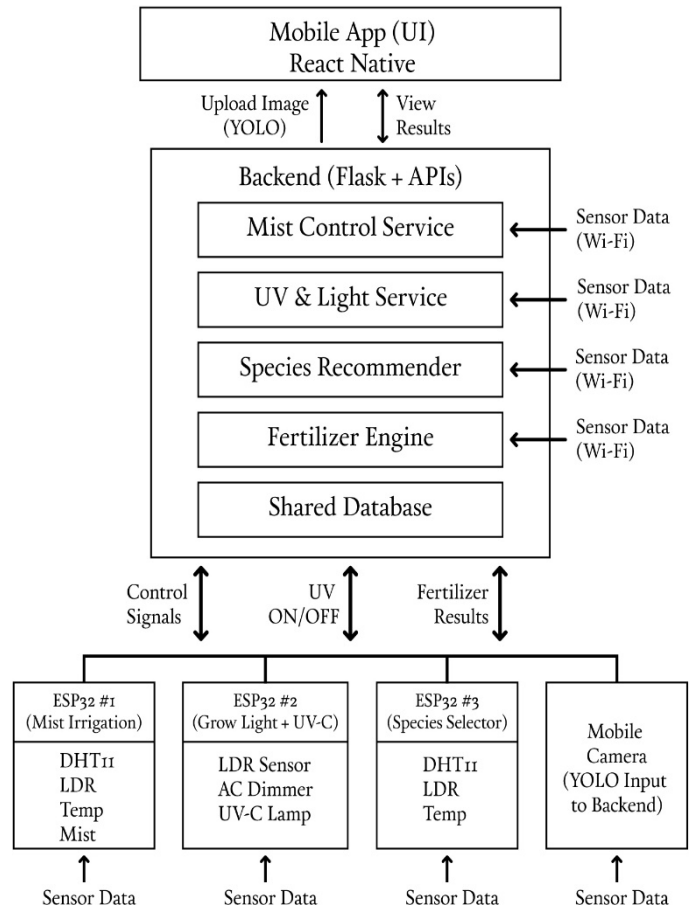


Fig. 1. High Level System Diagram.

V. RESULTS AND DISCUSSION

Advanced technologies of ML and IoT were incorporated and performed to maximize the growth factors of the orchid varieties of Dendrobium, Vanda, and Phalaenopsis. The created Orchid species recommendation system involved IoT environmental control and the ANN to determine specific growth conditions for the particular species of orchids. Other climatic factors such as humidity, temperature, and light intensity were simultaneously measured and quantified to consider variations in the experiment setting. This species data was then matched against a robust species database facilitating accurate matching of orchid species with their ideal growth conditions. The ANN model provided detection rates of 91%, 87%, and 90% for Phalaenopsis, Vanda, and Dendrobium, respectively. This approach improved growth rates, reduced compatibility risks, and increased productivity and profitability chances while in the meantime giving real-time suggestions on the best ways to alter plantation conditions.

The mist irrigation system utilized real-time of environmental data and programmed algorithmic control to manage the mist level to suit the water needs of the Dendrobium, Vanda, and Phalaenopsis types of orchids. Services related to the environmental context, including the relative humidity, temperature, and light intensity, were monitored by IoT-based sensors on a real-time basis. This data was used to control the timing of the system's mists so that the correct amount of water was distributed consistently without intervention. The thin layer of water used by overhead mist sprayers copied normal rainfall to facilitate root formation as well as to avoid water loss through transpiration on the foliage. Since the system was modular, there was the possibility of different irrigation schedules for different species which made the system very flexible. The use of water was conserved alongside the microclimate to provide appropriate conditions for the growth of orchids. Future improvements such as the incorporation of weather prediction will ensure the expansion of the scalability of the system for large-scale farming. The circuit organization can be shown below by Fig 2.



Fig. 2. Irrigation Circuit Organization.

The algorithm effectively ensures precise irrigation. The algorithm basic structure can be shown below by Fig 3.

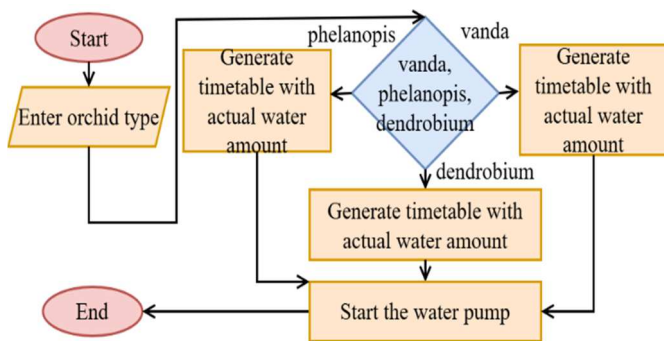


Fig. 3. Irrigation Algorithm basic structure.

The light optimization and UV treatment is a combination of ML and IoT can provide species-specific light and disease solutions. The study obtained 91%, 87%, and 90% detection rates for the three kinds of orchids in Phalaenopsis, Vanda, and Dendrobium, and the orchid diseases, using two YOLOv5-based ML models. The described lighting optimization system used LDR sensors and AC dimmers to control growing light intensity dividing it into three classes: low, medium, and high. The system adapted the light intensity to the current lighting situation to provide adequate lighting for each type of orchid while using on average 25% less energy than traditional methods. The confusion matrix of the YOLOv5 model is shown in Fig 4. For disease management, the UV-C treatment at 30 J/cm² was provided to the fungal infection areas identified by the YOLOv5 model. The UV was used in a pulse form which may not stress the plant while still having the capability to control the fungal pathogen.

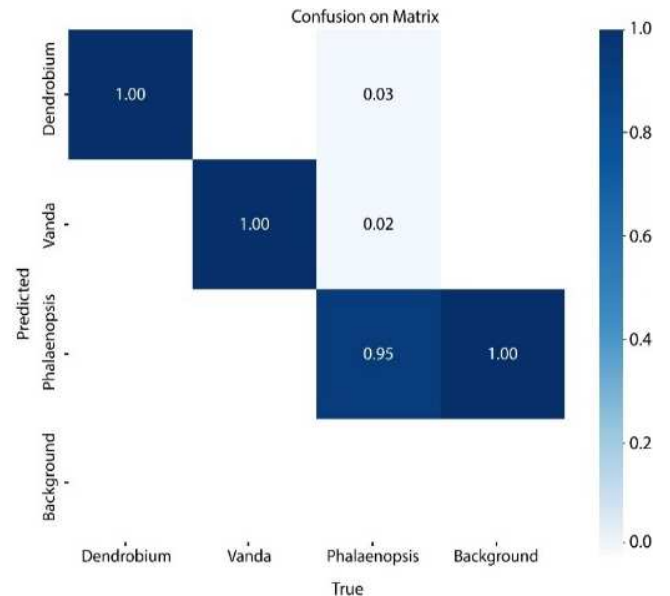


Fig. 4. Confusion Matrix For Orchid Type Identifier Model.

Studies confirmed that the UV does not harm plants at this dose level [13]. The overall evaluations of post-treatment revealed a 70 percent increase in infected plants after one cycle. Fungicides were suggested to be used alongside UV treatments to enhance plant healing. In the sequence of the UV treatment process, individuals and other plants around the UV safety box were protected from dangerous rays. This comprehensive approach demonstrated the system's capability to address both environmental light optimization and disease management effectively. The Disease preview is shown in Table 1.

TABLE I. ORCHIDE DISEASE PREVIEW

Disease	Fungus	Fungicide
Phytophthora	Phytophthora Parasitica	Subdue (metalaxyl)
Anthracoze	Colletotrichum Gloeosporioides	Daconil
Fusarium	Fusarium Oxysporum f.	Prosaro

Fertilizer Recommendation system which contains custom trained YOLO model performed effectively in finding plant features such as flowers, leaves, stems, and pots with high confidence; excelling in the leaf detection with a confidence level above 0.85, and flowers with a confidence of approximately 0.90. stems confidence level around 0.76 and pot categories, and also some misclassification of roots. The image for object detection is shown in Fig 5.

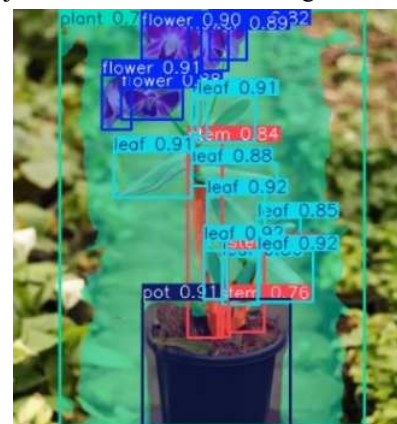


Fig. 5. Object detection of custom train YOLOv11.

In the prediction of age gaps from the model comparison, Logistic Regression was found to be inadequate with a minimum accuracy of 27.10% because of the inherent linearity in the model. On the other hand, K-Nearest Neighbors and Decision Tree both of have results with an equal accuracy score of 97.51%, which indicates that both are good at capturing complex relationships in the data. Though the two leading had equal accuracy, Decision Trees are recommended for better interpretability and faster predictions. This model is used to predict age gap. The confusion matrix for orchid age gap identification is shown in Fig 6.

Accuracy on test data: 0.9751

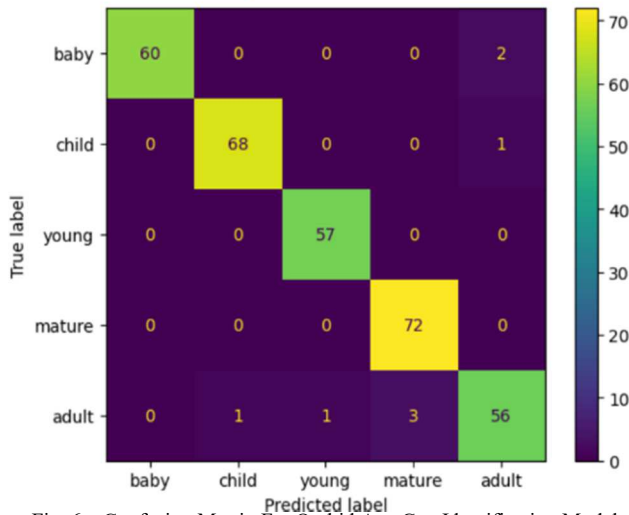


Fig. 6. Confusion Matrix For Orchid Age Gap Identification Model.

The smart farming system was validated over a period of four weeks using 60 orchids (*Dendrobium*, *Vanda*, *Phalaenopsis*), half of which received traditional care while the other half were under automated care. A mist irrigation system also reduced water usage by 32%, with humidity levels in the system ranging from 60 to 75%. The UV-C disease treatment, triggered using YOLOv5 detection, led to a 72% reduction in fungal symptoms within two cycles. Light optimization through LDR sensors and AC dimmers reduced energy consumption by 27% while improving flowering by 68%. Fertilizer recommendations based on both species and the growth stage of the plant increased plant health and flowering by 68% compared to controls. These data prove the effectiveness of the system on the sustainability, efficiency, and vitality of plants. OrchiZen is Sri Lanka's first and only intelligent orchid farming solution, specially made for orchid cultivation. Unlike the global platforms, it combines significant features specialized for orchids, thus offering the state-of-the-art floriculture to a local and commercial grower.

VI. CONCLUSION

OrchiZen is based on ML, IoT with advanced software engineering technologies and aims to increase the efficiency of growing orchids in including watering, selection of species, and eradication of diseases, as well as optimal nutrition. The smart irrigation system will guarantee water efficiency and best environmental conditions regarding the misting unit control. The species recommendation module helps in decision-making by relating environmental factors and orchid needs. UV-C light disease control and light optimization improve yields and health while reducing resource use. The recommendation system of fertilizer ensures that the nutrient

needs to be delivered for each of the growth phases. Collectively, these modules offer an end-to-end, integrated, and sustainable system solution that significantly raises the effectiveness of the orchid farm and increases eco-effectiveness for directing progression towards advanced smart agriculture.

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